

CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

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Abstract. University counseling spans relational curricula, cohort-specific fees, exemption policies, and narrative regulations, each requiring distinct evidence representations and computational treatments. Controlled comparisons against functionally matched single-agent alternatives remain limited. We present CTU-Chat, a supervisor-routed multi-agent system that assigns queries to four bounded specialists with scoped tool privileges and representation-specific evidence paths: Neo4j graph traversal for curricula, structured lookup with deterministic calculation for financial data, and hybrid document retrieval with neural reranking for regulations. We evaluate CTU-Chat through three experimental scenarios on a 100-query held-out benchmark: architecture comparison, five-component ablation, and semantic-neighbor tool stress testing up to 51 tools. In architecture comparison, CTU-Chat obtains an 8.0 pp higher End-to-End Success point estimate (81.0% vs. 73.0%; paired 95% CI [0.0, +16.0] pp; McNemar $p=0.096$) and 41.4% fewer input tokens at the cost of higher latency ($\sim 2\times$). Under registry expansion to 51 tools, scoped specialization preserves 88.0% E2E execution (vs. 78.7% for Top-10 selection), without displaying statistically divergent degradation slopes. These findings suggest that domain-scoped specialization can support institutional counseling reliability under the evaluated conditions, with measurable coordination trade-offs.

Keywords: Multi-Agent Systems · Supervisor Routing · Architectural Ablation · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

1 Introduction

Can Tho University (CTU) serves nearly 49,000 undergraduates in 2025–2026 [1]. Its scale and evolving regulations create advisory needs spanning heterogeneous knowledge forms: (1) **academic curricula** encoding hierarchies and prerequisites; (2) **tuition schedules** varying by cohort and field; (3) **exemption and scholarship policies** combining

eligibility conditions with rates; and (4) **administrative regulations** comprising lengthy narrative documents. Prior systems address individual domains—CAAS for IT admission counseling [10] and REBot for regulation advising [9]—but a single representation or retrieval method may be unsuitable across all domains.

RAG and GraphRAG augment LLMs with documents and knowledge graphs [8,5,3], while agentic architectures support tool use and multi-agent coordination [20,18,16]. These approaches do not prescribe how heterogeneous institutional evidence should be represented, routed, and processed. Known limitations in language-model arithmetic [14] further motivate deterministic calculation for financial queries.

Central hypothesis. When a counseling system must handle diverse knowledge forms and semantically adjacent tools, routing queries to bounded specialists with representation-specific evidence paths may improve reliability relative to a single agent with equivalent capabilities—but at the cost of coordination latency. Whether this trade-off is favorable requires direct comparative evaluation.

These challenges motivate three gaps: (1) limited evidence comparing multi-agent and single-agent architectures under matched capabilities; (2) limited mechanistic understanding of which components drive observed differences; and (3) limited evidence on how bounded specialization maintains reliability as the tool registry grows. We formulate three research questions:

- **RQ1:** How does the proposed multi-agent architecture compare with a functionally matched single-agent in end-to-end reliability, evidence grounding, and cost?
- **RQ2:** What performance changes are observed when specialist policies, tool isolation, representation-specific evidence access, deterministic computation, and route repair are individually removed or replaced?
- **RQ3:** How does bounded specialist routing affect tool-selection reliability as semantically overlapping tools expand the registry from 11 to 51 tools?

This work provides three contributions: **C1**—a supervisor-routed bounded multi-agent architecture with four specialists, deterministic route repair, and partitioned tool privileges; **C2**—an architecture-level comparative evaluation against a functionally matched single-agent baseline and a routed generic-worker variant; **C3**—architectural ablations of five components and a semantic-neighbor tool-registry stress test across 11–51 tools.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the CTU-Chat architecture. Section 4 describes the experimental setup. Section 5 reports results. Section 6 discusses findings and limitations. Section 7 concludes.

2 Related Work

2.1 Vietnamese NLP and Retrieval Foundations

Vietnamese is an isolating, tonal language where white spaces separate syllables rather than words [11]; 85% of word types contain at least two syllables. While domain-adapted bi-encoders capture semantic paraphrasing, Vietnamese higher-education decrees rely

on rigid Sino-Vietnamese legal terminology and alphanumeric cohort codes, motivating coupling dense embeddings with exact lexical matching.

BM25 [15] provides reliable exact token matching for rigid identifiers but cannot capture paraphrases. Dense retrieval [7,8] captures paraphrastic intent but may struggle with rare alphanumeric tokens. Reciprocal Rank Fusion (RRF) [2] merges BM25 and dense candidates without score calibration, while neural cross-encoders [12,19] jointly encode query–passage pairs for reranking. Adaptive-RAG [6] demonstrates complexity-based retrieval selection. These approaches motivate dynamic retrieval but do not allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

2.2 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples [13]. GraphRAG [3] leverages graph topology to uncover multi-hop associative paths that escape flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction may introduce spurious relationships, motivating selective graph construction restricted to domains with well-defined relational schemas.

2.3 Agentic RAG and Multi-Agent Orchestration

ReAct [20] demonstrates interleaved reasoning and action, AutoGen [18] provides a multi-agent conversation framework, and Toolformer [16] studies tool invocation. For institutional counseling, these capabilities raise design risks without explicit constraints: broad tool exposure increases errors; peer-to-peer exchanges increase coordination cost; a shared index mixes evidence across domains. Importantly, controlled comparisons against functionally matched single-agent baselines remain limited, as most evaluations report absolute performance without isolating the contribution of multi-agent decomposition from underlying capabilities.

2.4 Educational Counseling Systems

Educational chatbots assist in advising and learning support [17]. CAAS supports admission counseling for IT programs [10], while REBot combines retrieval with category-guided GraphRAG for academic regulations [9]. CTU-CHAT departs from prior systems across three aspects: (1) *Task scope*: modeling 113 curricula as relational property graphs with deterministic arithmetic for tuition; (2) *Multi-agent routing*: a LangGraph supervisor routes to four bounded specialists with isolated evidence spaces; and (3) *Architectural evaluation*: direct comparison against functionally matched baselines with component-level ablations.

2.5 Research Gaps

Synthesizing the reviewed approaches identifies three gaps: (1) limited architecture-level evidence comparing multi-agent and single-agent systems under matched capabilities;

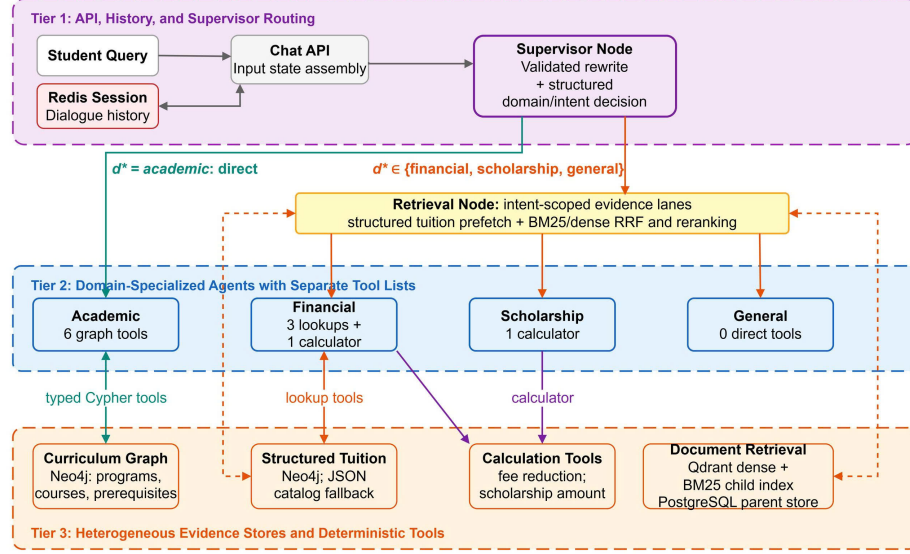


Fig. 1. Implemented CTU-CHAT architecture. Redis history enters the graph state; the supervisor’s structured decision is contract-checked before conditional dispatch. Academic queries enter the graph-enabled agent; other domains pass through intent-scoped retrieval.

(2) limited mechanistic understanding of which components drive observed differences; and (3) limited robustness evidence under tool ambiguity as semantic neighbors accumulate. These gaps motivate the evaluation design in Sections 4–5.

3 Proposed Model

This section presents CTU-CHAT’s architecture, algorithms, and evidence-acquisition mechanisms. The design follows three principles: *centralized agent coordination* (a contract-checked supervisor dispatches each query to one domain agent); *heterogeneous knowledge allocation* (each information structure maps to a corresponding evidence representation); and *representation-aware evidence routing* (graph-tool or intent-scoped retrieval path according to the assigned domain).

3.1 Overall Architecture

Figure 1 illustrates three coordinated tiers: (1) the Interaction and Centralized Supervisor Tier, (2) the Domain-Specialized Multi-Agent Core, and (3) the Heterogeneous Knowledge Storage and Tool Tier.

CTU-CHAT specializes LangGraph through typed state (`AgentState` with validated rewrite, routes, evidence, and response), an intent-owner contract that checks structured supervisor output before dispatch, conditional routing (academic queries go directly to

the graph agent; others through retrieval), acyclic orchestration (specialists return to END without peer edges), and asymmetric tool registries (each specialist sees only its scoped tools).

3.2 Neo4j Knowledge Graph and Cypher Tools

The Neo4j graph contains over 1,500 nodes and 3,800 edges across 113 undergraduate curricula and fee schedules. Its 10 node labels cover curriculum entities (textttProgram, Course, Block) and tuition entities (TuitionFee, TuitionPolicy, ExemptionBasisRate); 13 typed relationships encode containment, prerequisites, program outcomes, tracks, occupations, tuition, and exemption bases.

The system does not ask the LLM to generate arbitrary Cypher. After the supervisor selects the academic agent, its ReAct policy chooses one of six typed tools and supplies arguments. Each tool normalizes identifiers and calls a fixed, parameterized Cypher query implementing three query classes: *attribute lookup/search* (program codes, names, comparisons), *structural traversal* (prerequisite chains up to ten hops, course intersection, reverse path search), and *structured financial lookup* (tuition, policies, exemption rates). This design binds agent actions to deterministic graph operations, reducing exposure to errors from unconstrained Cypher generation.

3.3 Routing and Deterministic Arithmetic

For a query q and its dialogue predecessor, the supervisor first attempts a validated standalone rewrite, then predicts domain d_{raw} and intent ι_{raw} under a structured schema, falling back to *general/other* on parsing failure. The ownership contract repairs rule conflicts or incompatible domain–intent pairs. Academic queries proceed directly to the graph agent; other domains build intent-scoped retrieval lanes, with financial queries first attempting Neo4j tuition lookup and then JSON fallback.

Deterministic Tuition Arithmetic. For fee-reduction requests, the external tool `tinh_toan_hoc_phi` receives three parameters—per-credit tuition T , per-credit exemption basis B , and percentage p —and applies:

$$D = B \frac{p}{100}, \quad P = \max(0, T - D). \quad (1)$$

The tool enforces $0 \leq p \leq 100$ and clamps P at zero. The contribution lies in composing validated rewriting, constrained routing, and deterministic arithmetic within the multi-agent workflow.

3.4 Heterogeneous Knowledge Allocation and Hybrid Retrieval

Rather than a uniform document index, CTU-CHAT preserves relational curricula in Neo4j, accesses cohort-dependent tuition through structured lookup, and retains narrative regulations in document retrieval. Academic queries enter the graph agent directly; others enter retrieval. Financial retrieval first attempts Neo4j tuition lookup with JSON fallback; the repaired intent then selects metadata-filtered document lanes.

For narrative regulations and scholarship criteria, BM25 and Qdrant search 400-character child chunks; parent sections (2,800-char target, 100-char overlap) are loaded from PostgreSQL and fused via RRF: $s_{\text{RRF}}(d) = \sum_{s \in \{\text{bm25}, \text{dense}\}} (60 + r_s(d))^{-1}$. The full stack pools intent-governed and unfiltered candidates, deduplicates, and reranks with `bge-reranker-v2-m3`. Scores below 0.20 are removed unless fewer than two candidates remain; graph evidence is added under fixed quotas and up to seven passages are packed for generation.

3.5 Cross-Domain Query Handling

A bounded specialist architecture raises the question of how queries spanning multiple domains are handled. CTU-CHAT adopts a *single-owner with multi-evidence prefetch* policy: the supervisor selects one primary owner based on dominant intent, but the retrieval layer collects evidence from all relevant sources before passing aggregated context to the owner. Specialists do not invoke other specialists; if the query requires a tool outside the owner’s scope, evidence is prefetched through retrieval rather than cross-specialist delegation. This design trades full cross-domain tool composability for bounded execution. The 20 cross-domain held-out queries directly test this mechanism.

4 Experimental Setup

4.1 Dataset and Annotation

We evaluate CTU-CHAT on 200 administrative queries constructed through a document-grounded protocol over official CTU statutes. The 100-query development split supports index calibration and prompt engineering. The 100-query held-out set follows a predefined distribution: 40 direct, 20 within-domain multi-hop, 20 cross-domain, 10 entity-comparison, 5 temporal, and 5 adversarial queries (51 formal, 49 colloquial; 56 single-source, 44 multi-source). It contains 19 Academic, 28 Financial, 17 Scholarship, and 16 General single-domain queries; the remaining 20 combine evidence across domains. Candidate queries and ground-truth evidence were extracted from official statutes: one annotator established reference answers, required facts, and gold sources, while a second annotator independently verified all annotations; disagreements were adjudicated against official regulatory text. An audit found no cross-split duplicates (max 5-gram Jaccard = 0.1875). Each query is annotated along nine dimensions, including domain, evidence type, and tool requirements, to support fine-grained failure analysis.

4.2 Baselines and Configurations

Fairness principle and execution contract. All configurations share the same LLM (Gemini 2.5 Flash Lite, $T=0.0$), knowledge representations, retrieval stack (BM25 + dense + RRF + `bge-reranker-v2-m3`, $k=7$), and deterministic calculator access. Differences are strictly restricted to *orchestration architecture*. To isolate orchestration differences from retrieval variance in Scenario 1, all three architecture variants evaluate over the identical shared full-stack evidence retrieval output (T_4), yielding matched pre-generation

Source Recall (78.6%) and Source AP (70.8%). Performance divergences in final answers are therefore attributable purely to tool privilege scoping, prompt specialization, and agent routing. Multi-agent variants use intent-routed scoped registries with identical execution bounds: maximum 3 tool calls, matching context limits, 2,048 output-token budget, and 30-second timeouts. Token accounting for CTU-CHAT comprehensively aggregates all pipeline invocations (supervisor routing, contract repair, worker generation, and retries), ensuring rigorous parity with the single-agent baseline.

Scenario 1 — Architecture Comparison (RQ1): Three configurations are evaluated on the held-out set ($N=100$ queries, 300 runs under greedy decoding $T=0.0$): **S1-A:** Unified Single Agent with all 11 tools, no supervisor or routing; **S1-B:** Routed Generic Worker—supervisor routes to domain workers with scoped tools but a single generic prompt; **S1-C:** Full CTU-Chat with supervisor, contract repair, specialist prompts, scoped registries, and deterministic computation.

Scenario 2 — Ablation (RQ2): Five ablations each remove one component from Full CTU-Chat ($N=100$ queries per ablation, $T=0.0$): **S2-A1:** No Specialist Policy (generic worker prompt); **S2-A2:** No Tool Isolation (all 11 tools visible); **S2-A3:** Uniform Text-RAG Baseline (graph data serialized to text, traversal disabled); **S2-A4:** No Deterministic Calculation (LLM generates arithmetic directly); **S2-A5:** No Route Repair (raw supervisor output used directly).

Scenario 3 — Tool Ambiguity (RQ3): Fifty fixed cases are evaluated while 40 semantic-neighbor tools expand registries to 11, 31, and 51 tools. Four configurations are compared: Full-Registry Single Agent, Global Top-10 Single Agent, Routed Generic, and Routed Specialist. Each case uses 3 counterbalanced tool-order blocks, yielding $50 \times 4 \times 3 \times 3 = 1,800$ decisions across counterbalanced blocks.

4.3 Evaluation Metrics

Metrics span four groups. *Final-answer quality:* Required-Fact Coverage (continuous proportion of reference facts verified in generated response), Answer Correctness, Answer Relevancy, and Faithfulness (Ragas [4]). *Evidence quality:* Source Recall, Source AP, Context Recall, Context Precision. *Execution reliability:* Tool Invocations per Query, Tool Selection Accuracy, Argument EM/Contract Match, Result Accuracy, and route repair rate. *Operational cost:* input/output tokens, LLM call counts, and wall-clock latency. Six primary endpoints are reported in the comparative evaluation: E2E Success Rate, Required-Fact Coverage, Source Recall, Source AP, Input Tokens, and Median Latency.

E2E Success definition. For retrieval queries, E2E Success is evaluated under the criterion of required facts $\geq 50\%$, absence of unsupported claims, and retrieval of verified sources, complemented by continuous Fact Coverage. For tool queries: correct route, tool, arguments, result, and answer. For adversarial queries: absence of unnecessary tool calls and appropriate clarification or refusal. For the arithmetic subset ($N=9$), numeric exact match against official bursar schedules is independently audited.

4.4 Statistical Protocol

Evaluations use greedy decoding ($T=0.0$) to minimize sampling variance. Paired differences across configurations are computed at the query level ($N=100$); 95% CIs are estimated using 10,000 bootstrap resamples. Three primary pairwise comparisons (Scenario 1) are declared in advance; additional pairwise tests use Holm correction. Scenario 3 degradation is computed as $\Delta = \text{score}_{51} - \text{score}_{11}$ across counterbalanced tool-order blocks.

4.5 Infrastructure

Gemini 2.5 Flash Lite via Vertex AI ($T=0.0$). Neural reranking on a GTX 1650 (evaluation hardware). Neo4j 5.20 stores relational curricula; PostgreSQL stores parent chunks (2,800-char, 100 overlap); Qdrant indexes child chunks (400-char, 50 overlap) with a 768-dimensional vietnamese-bi-encoder. System uses `bge-reranker-v2-m3`, LangGraph StateGraph, and Redis 7. Latency excludes the first warm-up run. Prompt templates, tool contracts, and graph schema are archived.

5 Results

5.1 Scenario 1 — Architecture-Level Comparison

Table 1 compares the three configurations on the 100-query held-out benchmark. Because evidence retrieval is held fixed across configurations to strictly isolate orchestration effects, Source Recall (78.6%) and Source AP (70.8%) are identical across all three architectures. Differences in final answer quality and operational cost stem entirely from prompt specialization, tool privilege scoping, and agent routing.

Table 1. Scenario 1: Architecture-level comparison ($N = 100$). Tools/Q = mean tool invocations ($\downarrow$). Bold = best reliability / lowest cost.

Arch.	E2E	Fact	Src.R	Src.AP	Tools/Q	In Tok	Out Tok	Lat. (ms)
Single Agent	73.0	63.8	78.6	70.8	0.41	6296	1530	5159
Routed Gen.	71.0	59.6	78.6	70.8	0.31	3473	1415	8921
CTU-Chat	81.0	66.7	78.6	70.8	0.28	3690	2092	10652

Pairwise bootstrap analysis (10,000 resamples): (1) CTU-Chat vs. Single Agent: CTU-Chat achieves an 8.0pp higher E2E point estimate (81.0% vs. 73.0%, paired 95% bootstrap CI $[0.0, +16.0]$ pp; exact McNemar $p=0.096$), alongside higher Fact Coverage ($\Delta = +0.028$, 95% CI $[-0.021, +0.079]$) and a 41.4% reduction in input tokens ($\Delta = -2607$, CI $[-3325, -1868]$), at the cost of +5.49 s median end-to-end latency. (2) CTU-Chat vs. Routed Generic: specialist policies raise Fact Coverage by +7.05 pp (CI $[+0.010, +0.132]$, $p < 0.05$). (3) Routed Generic vs. Single Agent: comparable token

savings ($\Delta = -2824$) but lower fact coverage ($\Delta = -0.042$, CI $[-0.098, +0.013]$), indicating that routing without policy specialization is insufficient.

On the 20 cross-domain queries, CTU-Chat achieves an 85.0% E2E point estimate compared with 75.0% (Single) and 70.0% (Generic), consistent with the intended single-owner multi-evidence-prefetch design.

5.2 Scenario 2 — Architectural Ablation

Table 2 removes one mechanism at a time from Full CTU-Chat ($N=100$). Deltas are relative to the full system.

Table 2. Scenario 2: Ablation results ($N = 100$). Negative Δ = degradation.

Ablated	Δ E2E	Δ Fact	Δ Src.R	Δ Tok	Δ Lat
No Spec. Policy	-10.0	-7.05	0.0	-217	-1731
Full Tool Vis.	-3.0	-0.02	0.0	+1270	+296
Uniform Text-RAG	-10.0	-2.67	-12.3	-300	-1910
No Route Rep.	-4.0	-2.05	0.0	-219	-1610
No Determ. Calc. [†]	-6.0	-0.97	0.0	-782	-4540

[†]Full benchmark with calculators disabled. Δ Tok/Lat in tokens/ms per query.

The two largest observed degradations occur in *No Specialist Policy* (Δ E2E = -10.0 pp, 95% CI $[-19.0, -2.0]$ pp; Δ Fact = -7.05 pp, CI $[-13.13, -1.22]$ pp) and *Uniform Text-RAG Baseline* (Δ E2E = -10.0 pp, CI $[-16.0, -5.0]$ pp; Δ Src.R = -12.3 pp, CI $[-16.5, -8.3]$ pp). Removing tool isolation causes a minor shift (Δ E2E = -3.0 pp, CI $[-9.0, +3.0]$ pp, crossing zero) while expanding input tokens by +1,270. Disabling route repair yields Δ E2E = -4.0 pp (CI $[-10.0, 0.0]$ pp), disproportionately harming cross-domain queries (-15.0 pp). Disabling deterministic calculation produces Δ E2E = -6.0 pp (CI $[-12.0, -1.0]$ pp) across the benchmark; on the arithmetic tuition subset ($N=9$), exact-match answers drop from 8/9 to 3/9 (-55.6 pp exact match pass) due to unconstrained arithmetic hallucination.

Table 3. Scenario 2: E2E Success by query family across ablations.

Config.	Graph ($n=19$)	Struct-fin ($n=28$)	Narr-reg ($n=33$)	Cross ($n=20$)
Full CTU-Chat	73.7	85.7	78.8	85.0
No Spec. Policy	63.2	82.1	66.7	70.0
Full Tool Vis.	78.9	71.4	78.8	85.0
Uniform Text-RAG	73.7	67.9	75.8	65.0
No Route Rep.	78.9	78.6	75.8	70.0

5.3 Scenario 3 — Tool Ambiguity Stress Test

Table 4 reports the semantic-neighbor experiment across 11–51 tools ($N=50$, 3 counter-balanced tool-order blocks, 1,800 primary decisions).

Table 4. Scenario 3: Semantic-neighbor tool ambiguity ($N = 50$, $R = 3$).

Reg.	Config.	Vis.	Sel.	E2E	Tok	Lat
11	Full-reg single	11	94.0	86.0	1663	1098
	Top-10 single	10	93.3	83.3	1541	1148
	Routed gen.	5.1	90.7	80.0	1204	1889
	Routed spec.	5.1	93.3	85.3	1441	1983
31	Full-reg single	31	91.3	84.7	2706	1157
	Top-10 single	10	90.0	81.3	1149	1033
	Routed gen.	9.8	88.0	79.3	1396	1958
	Routed spec.	9.8	93.3	88.7	1830	2015
51	Full-reg single	51	92.7	83.3	3722	1289
	Top-10 single	10	86.7	78.7	1036	997
	Routed gen.	9.8	91.3	82.7	1340	1907
	Routed spec.	9.8	92.7	88.0	1770	2021

As the registry scales from 11 to 51 tools, the Full-Registry Single Agent degrades from 86.0% to 83.3% E2E while input tokens expand by 124%. The Global Top-10 drops to 78.7% (coinciding with shortlist truncation). The Routed Specialist preserves 88.0% E2E at 51 tools with only 9.8 visible tools, maintaining a fixed-registry endpoint advantage (+4.7 pp over Single Agent, +9.3 pp over Top-10). The degradation interaction (Specialist vs. Top-10 high-minus-low difference) is +7.3 pp (95% CI [0.0, +15.3] pp), confirming an endpoint advantage without demonstrating statistically divergent degradation slopes.

5.4 Supporting Retrieval Validation

Table 5 validates the retrieval subsystem independently.

Table 5. Supporting evaluation: Progressive retrieval component stacking ($N = 100$).

Config.	Hit@1	Hit@3	P@5	R@5	MRR@10	Lat. (ms)
BM25 Lexical	0.810	0.920	0.234	0.937	0.874	14
Dense Semantic	0.630	0.840	0.202	0.853	0.736	80
Hybrid RRF	0.770	0.940	0.222	0.935	0.851	80
Hybrid + Reranker	0.830	0.940	0.230	0.955	0.896	1,145
Full Retrieval	0.850	0.960	0.234	0.975	0.913	1,153

The progressive retrieval ablation confirms that each component contributes complementary strengths. BM25 provides a competitive baseline (0.810 Hit@1, 0.874 MRR@10) due to the frequency of rigid identifiers in Vietnamese administrative regulations (decree numbers, course codes). Dense semantic search exhibits lower recall (0.853 R@5) from vocabulary mismatch on specialized institutional nomenclature. Hybrid RRF fusion expands candidate breadth (0.940 Hit@3), though top-rank precision is slightly diluted by dense candidates relative to pure BM25. Incorporating cross-encoder neural reranking (`bge-reranker-v2-m3`) overcomes this dilution, surpassing the BM25 baseline (0.830 Hit@1, 0.896 MRR@10). Finally, graph grounding further lifts full-stack retrieval performance to 0.850 Hit@1 and 0.975 Recall@5, indicating that structured knowledge constraints effectively resolve institutional entity ambiguities.

5.5 Failure Analysis

Table 6 categorizes error modes across the 46 failure instances (< 50% Fact Coverage) in Scenario 1 (27 for Single Agent, 19 for CTU-Chat).

Table 6. Failure taxonomy across architecture configurations ($N = 100$). Categories are non-mutually exclusive; a single query may exhibit multiple error modes.

Category	Failure Type	Single CTU	
Routing	Misroute corrected by repair	N/A	4
	Unrecoverable misroute	N/A	2
Retrieval	Gold source absent	2	2
	Cross-domain partial omission	5	3
	Reranker demoted gold below $k=5$	4	4
Tool	Invalid/default arguments	4	1
	Wrong-family hallucination	3	1
	Unnecessary invocation	5	2
Generation	Tool result ignored in text	2	1
	Partial fact omission	11	9
Total	Fact Coverage < 50%	27	19

CTU-Chat’s advantage is concentrated in tool-execution reliability (2 tool errors vs. 7; 2 unnecessary calls vs. 5). Gold-source absence and reranker-demotion counts are identical (2 and 4 respectively), while CTU-Chat incurs fewer cross-domain partial omissions (3 vs. 5), confirming that retrieval quality is an orthogonal bottleneck.

6 Discussion

6.1 Answering the Research Questions

RQ1 (Architecture-Level Effectiveness): The architecture-level comparison indicates that supervisor-routed multi-agent decomposition obtains an 8.0 pp higher E2E point

estimate than the Unified Single Agent (81.0% vs. 73.0%), although the paired bootstrap interval touches zero ($[0.0, +16.0]$ pp; McNemar $p=0.096$). CTU-Chat also attains higher continuous Fact Coverage ($\Delta=+0.028$, 95% CI $[-0.021, +0.079]$) and 41.4% fewer input tokens (3,690 vs. 6,296), alongside a median latency trade-off (10,652 vs. 5,159 ms). The Routed Generic configuration (59.6% Fact Coverage) demonstrates that routing without specialist policies impairs performance ($\Delta=-0.071$ vs. CTU-Chat, $p<0.05$), indicating that multi-agent decomposition is beneficial only when paired with policy specialization.

RQ2 (Architectural Mechanism): The ablations support the hypothesis that the architectural components serve distinct functions: (1) Replacing representation-specific access with uniform text retrieval (S2-A3) produces the steepest Source Recall drop (-12.3 pp), with structured-financial queries suffering most (-17.8 pp); (2) Removing specialist prompts (S2-A1) causes the largest Fact Coverage drop (-7.05 pp); (3) Full tool visibility (S2-A2) inflates tokens by +1,270 without improving fact coverage, confirming tool isolation as a cost-control mechanism; (4) Omitting route repair (S2-A5) degrades cross-domain queries by -15.0 pp; (5) Disabling deterministic calculation (S2-A4) causes -6.0 pp E2E loss on the full benchmark and a 55.6 pp pass degradation on the arithmetic tuition subset.

RQ3 (Tool Ambiguity): As registries scale from 11 to 51 tools, the Routed Specialist preserves 88.0% E2E with 9.8 visible tools. The Full-Registry Single Agent degrades from 86.0% to 83.3% while expanding tokens by 124%; the Global Top-10 drops to 78.7% (coinciding with shortlist truncation). These observations indicate that scoped tool registries preserve endpoint execution reliability at 51 tools, though registry expansion did not induce statistically divergent degradation slopes.

6.2 Relation to Prior Work

Our findings connect to several active themes. While ReAct [20] and Toolformer [16] demonstrate tool execution in single-agent settings, our results show that unconstrained registries suffer token inflation and elevated tool confusion under semantic distractors. General multi-agent frameworks such as AutoGen [18] permit flexible inter-agent interaction; CTU-Chat instead adopts strictly hierarchical, contract-validated routing with single-owner prefetch. Our ablation (S2-A3) aligns with findings in GraphRAG [3], indicating that replacing representation-specific access with uniform text retrieval substantially reduces source recall in the evaluated benchmark. Unlike Adaptive-RAG [6], which routes by semantic complexity, CTU-Chat routes by institutional representation requirements.

6.3 Threats to Validity

Internal: Single LLM (Gemini 2.5 Flash Lite); while temperature-zero decoding minimizes sampling variance, minor API-serving non-determinism and prompt sensitivity remain potential confounds. **External:** Single-institution benchmark (100 queries, 5 temporal, 5 adversarial); results may not generalize to other regulatory structures. Semantic-neighbor tools are evaluation constructs. **Construct:** E2E metrics rely partly on Ragas automated evaluation; required-fact matching reduces generator-judge bias. Human evaluation on a subset would strengthen validity. **Statistical:** Bootstrap CIs

are query-level ($N=100$). The arithmetic ablation subset is small ($N=9$), though exact counts show 8/9 exact matches with calculator vs. 3/9 without. Three registry sizes do not support asymptotic scaling claims. **Operational:** Neural reranking was evaluated on dedicated GPU hardware. Wall-clock latency includes API and network serving variability and should therefore be interpreted as deployment-specific.

7 Conclusion and Future Work

This study developed and evaluated CTU-CHAT, a supervisor-routed multi-agent RAG framework for heterogeneous university counseling, through architecture-level comparison, mechanistic ablation, and tool-ambiguity stress testing.

Three principal findings address our research questions. For RQ1, supervisor-routed multi-agent decomposition obtains an 8.0 pp higher End-to-End Success point estimate (81.0% vs. 73.0%) and 41.4% fewer input tokens than a functionally matched single-agent baseline, with an associated latency trade-off. For RQ2, ablation reveals that specialist policies (−10.0 pp E2E) and heterogeneous evidence paths (−12.3 pp Source Recall) are the components associated with the largest observed degradations, with structured-financial and cross-domain queries degrading most under uniform text retrieval. For RQ3, the Routed Specialist maintains an endpoint execution advantage at 51 tools (88.0% vs. 83.3% for Single Agent and 78.7% for Global Top-10) without displaying statistically divergent degradation slopes.

These findings indicate that supervisor-routed bounded specialization—combining domain-specific policies, scoped tool access, representation-specific evidence paths, and deterministic calculation—supports institutional counseling reliability under the evaluated conditions, with measurable coordination trade-offs. Future work includes reducing routing overhead via parallel dispatch, extending benchmarks to bilingual settings, and conducting field trials on student advisory efficacy.

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